

1. Database Schema (MySQL/XAMPP)

```
CREATE DATABASE IF NOT EXISTS smart_waste_db;
USE smart_waste_db;

CREATE TABLE users (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
    role VARCHAR(50) NOT NULL DEFAULT 'citizen',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE waste_types (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) UNIQUE NOT NULL,
    description TEXT
);

CREATE TABLE wastes (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT,
    type VARCHAR(50) NOT NULL,
    latitude DECIMAL(9,6) NOT NULL,
    longitude DECIMAL(9,6) NOT NULL,
    image VARCHAR(255),
    status VARCHAR(50) DEFAULT 'pending',
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL
);

CREATE TABLE waste_collections (
    id INT AUTO_INCREMENT PRIMARY KEY,
    waste_id INT,
    worker_id INT,
    collection_time DATETIME,
    status VARCHAR(50) DEFAULT 'scheduled',
    notes TEXT,
    FOREIGN KEY (waste_id) REFERENCES wastes(id) ON DELETE CASCADE,
    FOREIGN KEY (worker_id) REFERENCES users(id) ON DELETE SET NULL
);
```

```
CREATE TABLE notifications (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT,  
  message TEXT NOT NULL,  
  is_read BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE  
);
```

2. Backend Code (Node.js + MySQL/XAMPP)

server.js

```
const express = require("express");  
const cors = require("cors");  
const path = require("path");  
  
const app = express();  
app.use(cors());  
app.use(express.json());  
app.use("/uploads", express.static(path.join(__dirname, "uploads")));  
  
app.use("/api/auth", require("./routes/auth"));  
app.use("/api/waste", require("./routes/waste"));  
app.use("/api/collection", require("./routes/collection"));  
app.use("/api/notifications", require("./routes/notifications"));  
  
const PORT = 5000;  
app.listen(PORT, () => console.log(`Server running on port ${PORT}`));
```

models/db.js

```
const mysql = require("mysql2");  
const pool = mysql.createPool({  
  host: "localhost",  
  user: "root",  
  password: "",  
  database: "smart_waste_db",  
  waitForConnections: true,  
  connectionLimit: 10,  
  queueLimit: 0  
});
```

```
});  
module.exports = pool.promise();
```

middleware/auth.js

```
const jwt = require("jsonwebtoken");  
const SECRET_KEY = "supersecretkey";  
const verifyToken = (req, res, next) => {...};  
const authorizeRoles = (...roles) => {...};  
module.exports = { verifyToken, authorizeRoles, SECRET_KEY };
```

routes/auth.js, waste.js, collection.js, notifications.js - Full code as provided above.

3. Frontend Code (Vite + React)

api/api.js

```
import axios from "axios";  
const API = axios.create({ baseURL: "http://localhost:5000/api" });  
export const setAuthToken = (token) => {...};  
export default API;
```

App.jsx

```
import { BrowserRouter as Router, Routes, Route, Navigate } from "react-router-dom";  
// Routes: Login, Register, Dashboard, ReportWaste, Collections, Notifications
```

Pages: - Register.jsx, Login.jsx, ReportWaste.jsx, Collections.jsx, Notifications.jsx (full code as above)

main.jsx remains default from Vite React template.

4. Usage Instructions

1. Run XAMPP, create database with above schema.
2. Start backend: `node server.js`.
3. Start frontend: `npm run dev` inside `smart-waste-frontend`.
4. Access frontend at `http://localhost:5173`.

5. Register users, login, report waste, view collections, check notifications.

End of Smart Waste Management System Project